Date: 18/03/2026

Duration: 1 hour 45

CONTINUOUS ASSESSMENT 1

II YEAR – SEMESTER 4

CSE23AE204 - PCP III

(B.Tech. E01,E02,E03,E05,E06, E61)

SET D

Question 1: Problem Statement

You are given a JSON object:

```
{
  "g": [g1, g2, g3, ...],
  "s": [s1, s2, s3, ...]
}
```

Where:

- g[i] represents the **greed factor** of the ith child.
- s[j] represents the **size of the jth cookie**.

A child can be satisfied if:

cookie_size >= greed_factor

Rules:

- Each child can receive **at most one cookie**.
- Each cookie can be assigned to **only one child**.

Your task is to **maximize the number of satisfied children**.

Output Format

Return:

satisfied_children

Where:

- satisfied_children → maximum number of children that can be satisfied.

Example 1

Input

```
{
  "g": [1,2,3],
  "s": [1,1]
}
```

Output

1

Explanation

Children greed factors:

[1,2,3]

Cookies:

[1,1]

Only one child with greed 1 can be satisfied.

Result:

1

Question 2: Problem Statement

You are given **two integer arrays**:

- gas[i] → amount of gas available at station i
- cost[i] → gas required to travel from station i to station i+1

All gas stations form a **circular route**.

If you start at station i, you:

1. Fill gas[i]
2. Travel to the next station using cost[i] fuel.

Your task is to determine the **starting gas station index** from which you can travel the **entire circular route exactly once**.

If completing the circuit is **not possible**, return **-1**.

Example 1

Input

gas = [1,2,3,4,5]

cost = [3,4,5,1,2]

Output

3

Explanation

Start at station **3**:

tank = 4

4 - 1 = 3 → +5 = 8

8 - 2 = 6 → +1 = 7

7 - 3 = 4 → +2 = 6

6 - 4 = 2 → +3 = 5

5 - 5 = 0 → +4 = 4

The circuit completes successfully.

Question 3: Problem Statement

A company has a network of **n servers**, numbered from **0 to n-1**.

The dependencies between servers are given as a list of directed edges:

edges = [[u1, v1], [u2, v2], ...]

Each edge [u, v] means:

- Server **u can activate server v**

Some servers depend on others, while some can operate independently.

Your task is to determine the **minimum set of servers that must be started manually** so that **all servers in the network become active**.

Return the list of such servers in **ascending order**.

Example 1

Input

n = 6

edges = [[0,1],[0,2],[2,5]]

Output

[0,3,4]

Explanation

0 → 1

0 → 2 → 5

3

4

Servers **3 and 4** are not reachable from any other node.

So we must start **0, 3, and 4**.